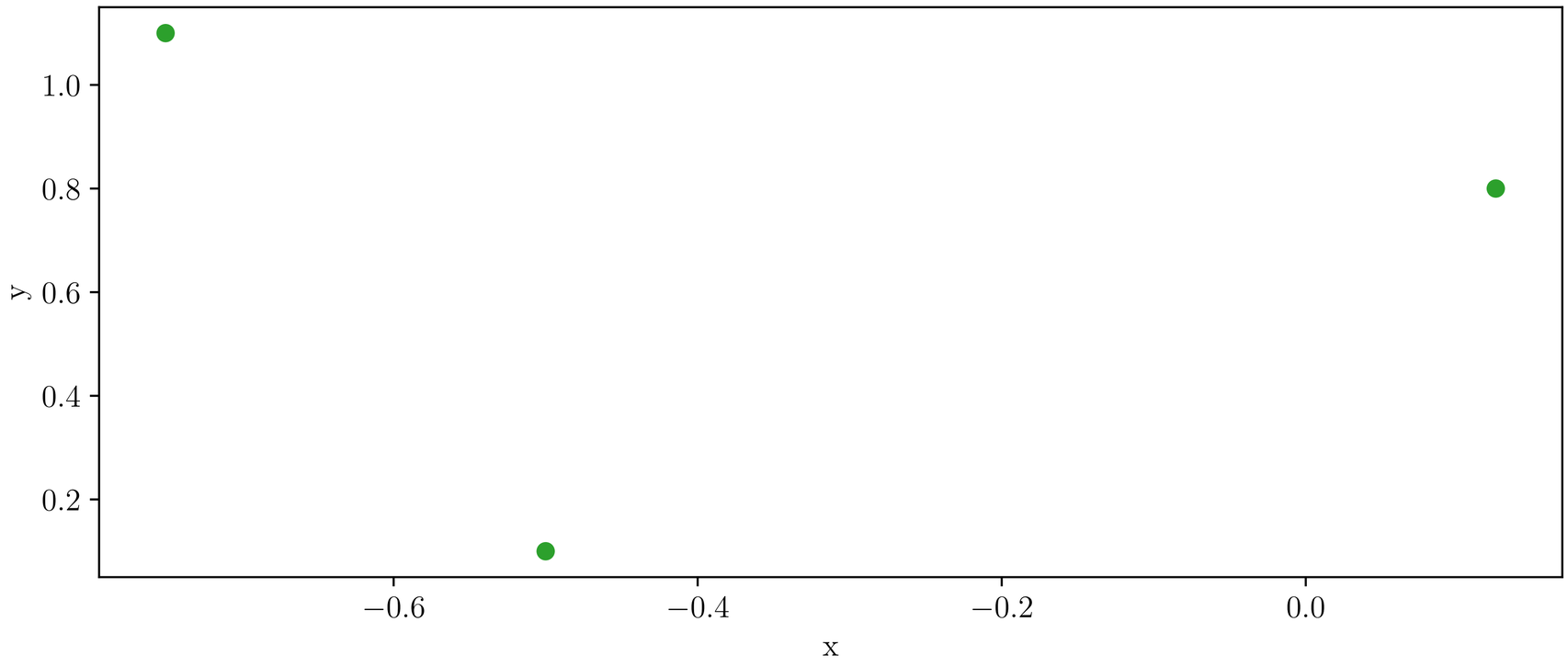
3 points example

Regression & ReLU again

3 points in $\mathbb{R}^2$

second coordinate = 1 (bias term)



Note:

- $\langle x_k, x_{k'} \rangle > 0$
- $y_k > 0$

Single (non-trivial) critical point of G given by

$$D = \frac{1}{n} \sum_k y_k x_k$$

Convergence to spurious minima

Theorem

If $\lambda \leq \lambda^*$, the gradient flow converges for any $m \in \mathbb{N}$ to some θ_∞ such that

$$h_{\theta_\infty}(x) = x^\top \beta^* \text{ for any } x$$

- $L(\theta_\infty) = \Omega(1)$
- **λ^* does not depend on m**
- holds for a set of data of non-zero measure (non-degenerate)